

EBERHARD KARLS
UNIVERSITÄT
TÜBINGEN



DATA LITERACY

MIDTERM EXAM FOR WINTER 2023/24
19.12.2023

Full name of student: _____

degree studied: _____
(e.g. MSc Machine Learning / BSc Medieninformatik / etc. If you are a PhD student, write "PhD" and, if applicable, the name of your graduate school)

Matrikelnummer (if applicable): _____

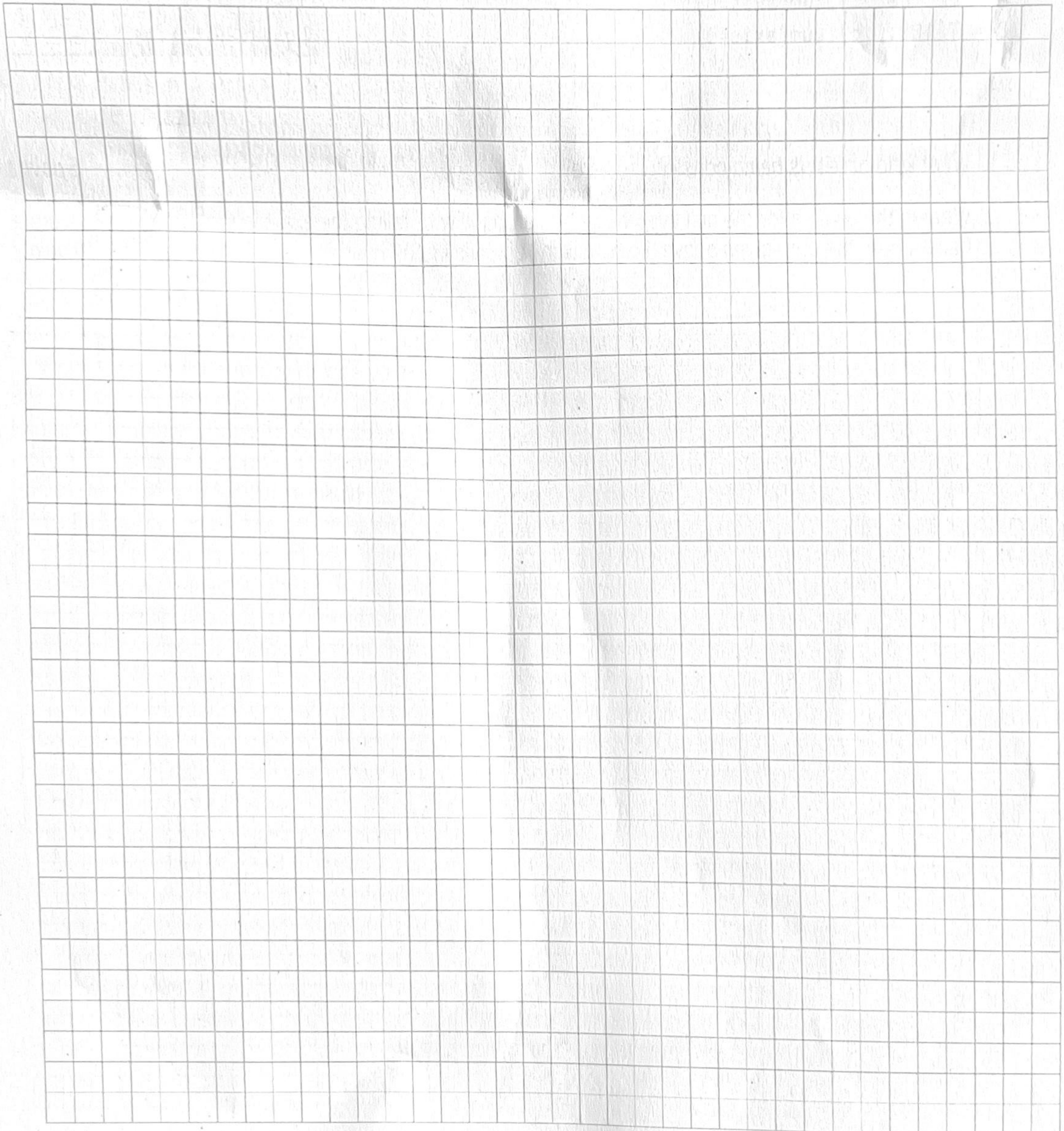
Before you begin, read the following:

- you have 60 minutes to complete the exam
- Points can be deducted for imprecise, insufficiently formal, or unreadable answers.
- there are 5 questions. Note that questions are printed on **both sides of the paper**.
- write your name, legibly, on **every** sheet of the exam
- write your answer directly below the question in the dedicated space on the top right.
If you need additional paper, there are two additional blank pages at the end. If those are not enough, ask one of the examiners for more paper. **Do not continue the answer for one question on the sheet for another question.** Do not separate the pages from each other.
If you are using additional paper, hand it in alongside your exam, it will be stapled to the back.
- multiple-choice questions can have several correct answers

Your Name: _____

Question 1 of 5: Bayes' theorem. As of 15 December 2023, the 7-day incidence of Covid19 in Tübingen county is assumed to be about 40 cases per 100,000 inhabitants (source: Robert Koch-Institut). If we assume for simplicity that a Covid infection lasts for about a week, then 0.04% of the population are currently symptomatic. Among infected people, about 10% experience headaches as a symptom. For simplicity, let us assume that the average healthy person experiences headaches about 1% of the time (so 3–4 days per year).

1. Write down Bayes' theorem. 4 points
2. One day this week, you wake up with a headache. Assuming the numbers quoted above are correct, what is the probability that you are infected with Covid19? 8 points
3. With a heavy head, you take a rapid antigen test, which comes back **positive**. Such tests have a **true positive rate** of 83% and a **false positive rate** of 1%. What is the probability that you have Covid now, given that you have both a headache and a positive test? (You may assume that headache and positive tests are conditionally independent events, given the presence or absence of Covid). 10 points

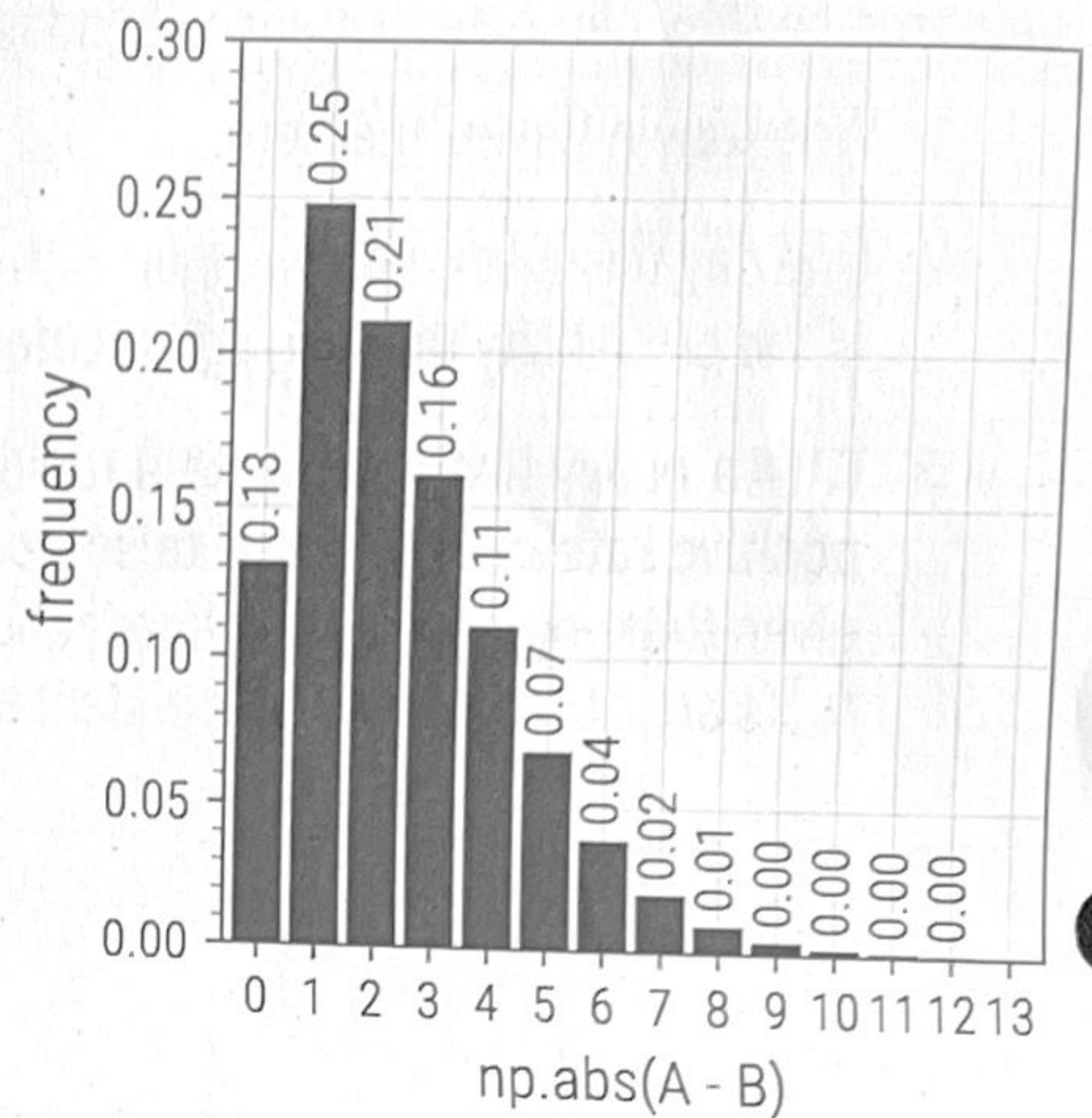
A large grid of graph paper, approximately 30 squares wide and 30 squares high, intended for students to show their work for the problems.

Your Name: _____

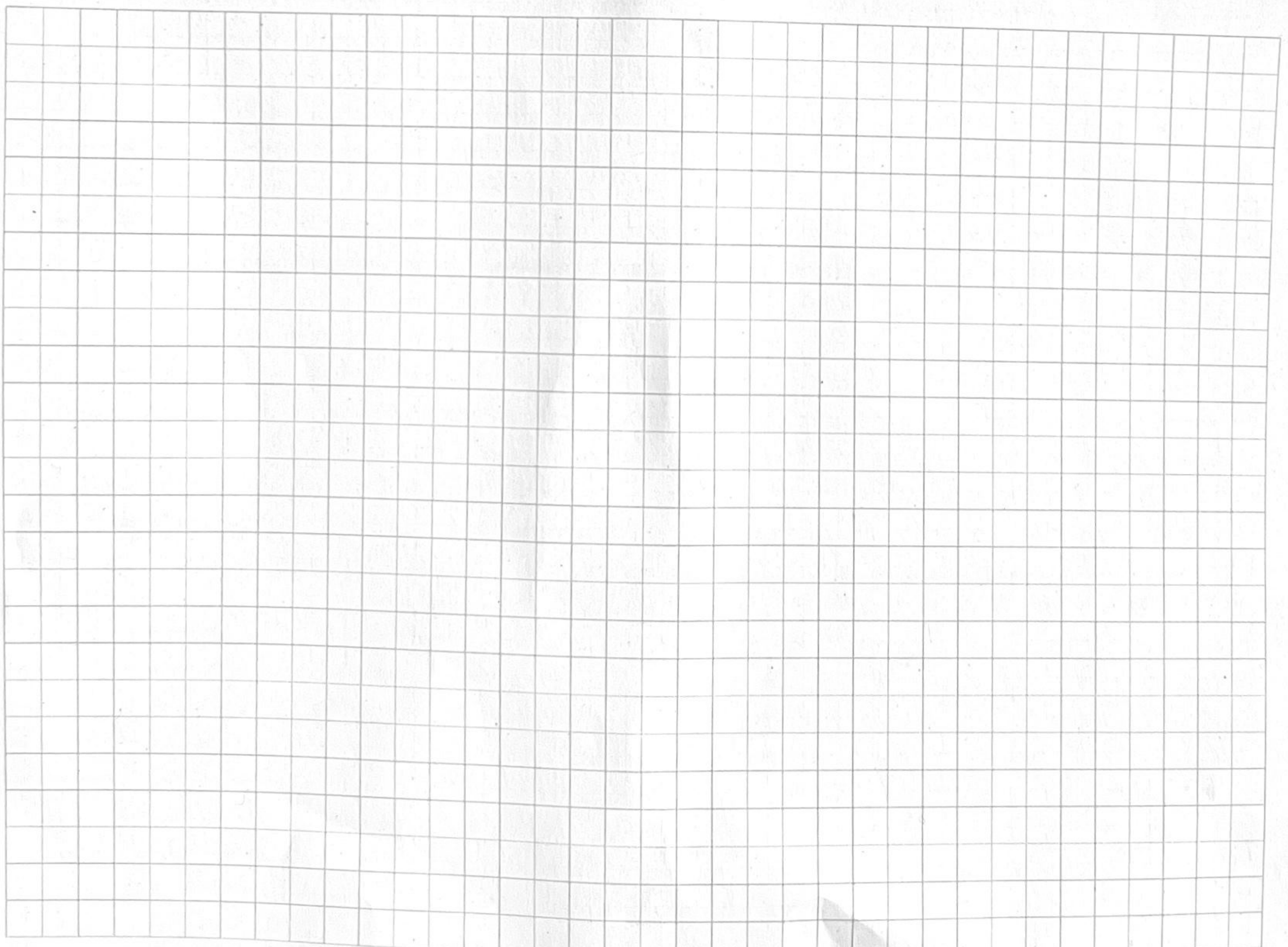
Question 2 of 5: An exam is being taken in parallel in two separate lecture halls A and B . $N = 200$ students participate in the exam, exactly 100 each in each lecture hall. 8 of the students from room A fail the exam, but only 2 of the students from room B fail. The plot shown below on the right shows a histogram that uses the output arrays A and B from the python code on the left.

```
import numpy as np
N = 200
K = np.zeros(N)
K[:10] = 1
S = int(1e7)
rng = np.random.default_rng(1)
T = rng.choice(K, size=(N, S), replace=True)

A = T[:100, :].sum(axis=0)
B = T[100:, :].sum(axis=0)
```



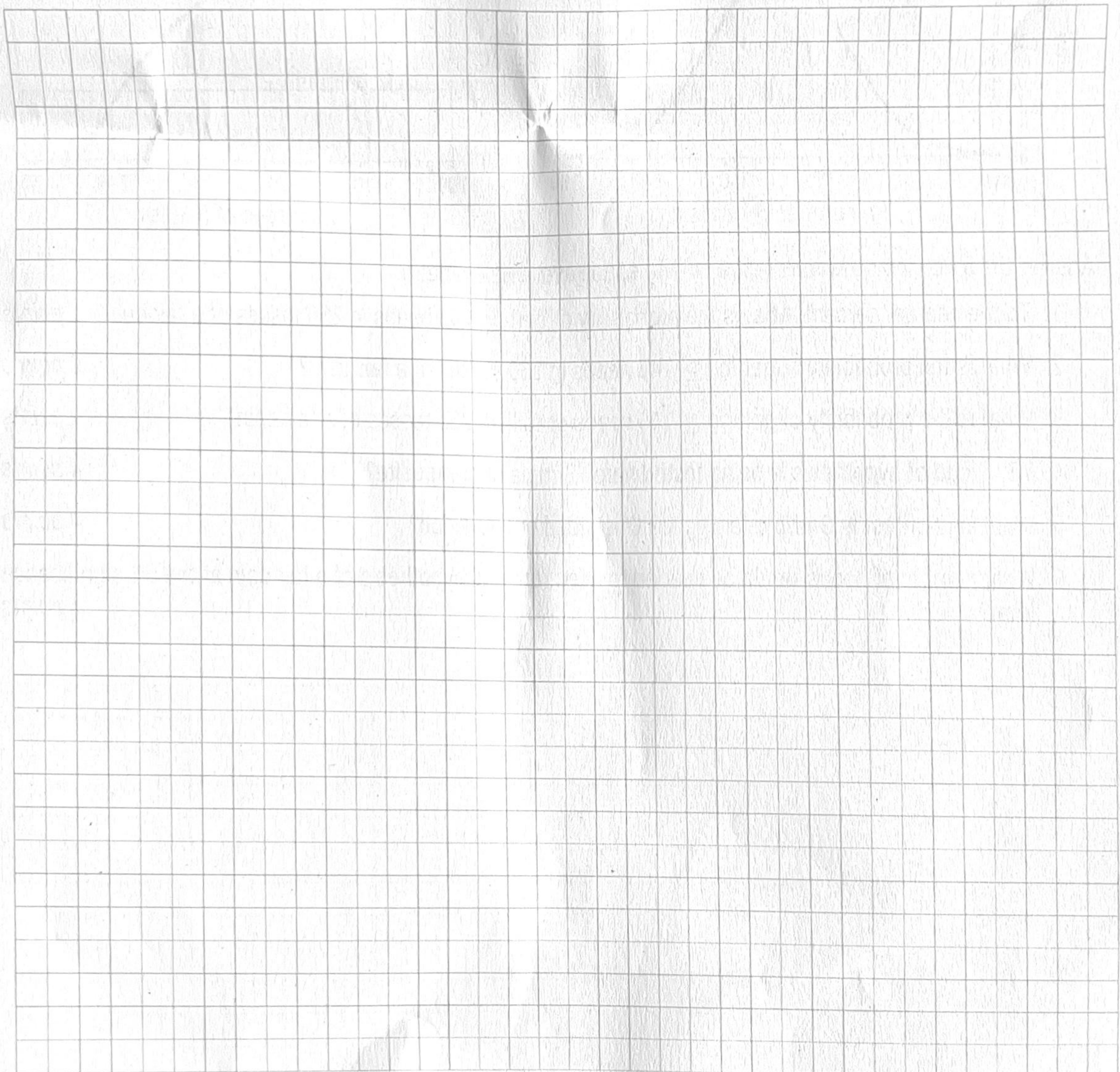
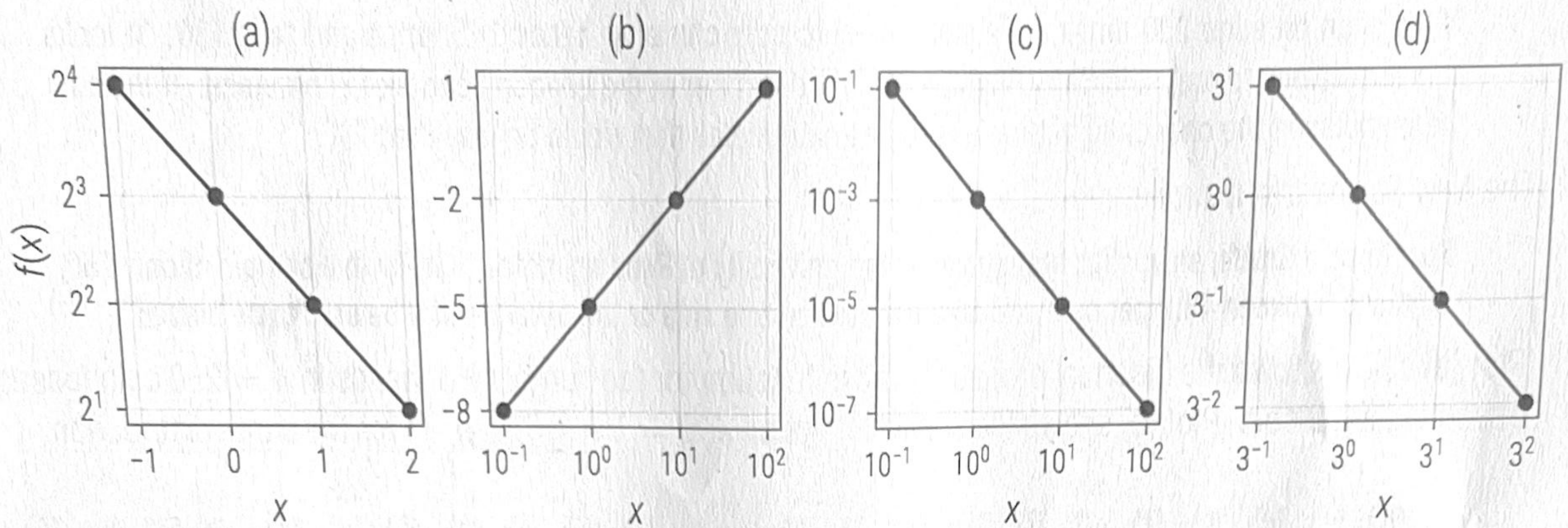
1. What kind of test is being constructed here?
5 points
2. What is the p -value for the null hypothesis "the probability of failing the exam is the same in both lecture halls" under the data given above (i.e. 8 fails in A , 2 fails in B)?
10 points



Your Name: _____

Question 3 of 5: *Plotting*. The panels below show four different functions. Note the different axis scales. For each function, deduce its algebraic form (i.e. write down the formula for the function). You can assume that the lines pass exactly through the grid at the marked points. Simplify all expressions as much as possible.

5 points each



Age (years)	Percentage (%)
18	10
20	15
25	25
30	40
35	55
40	70
45	80
50	85
55	88
60	90
65	90

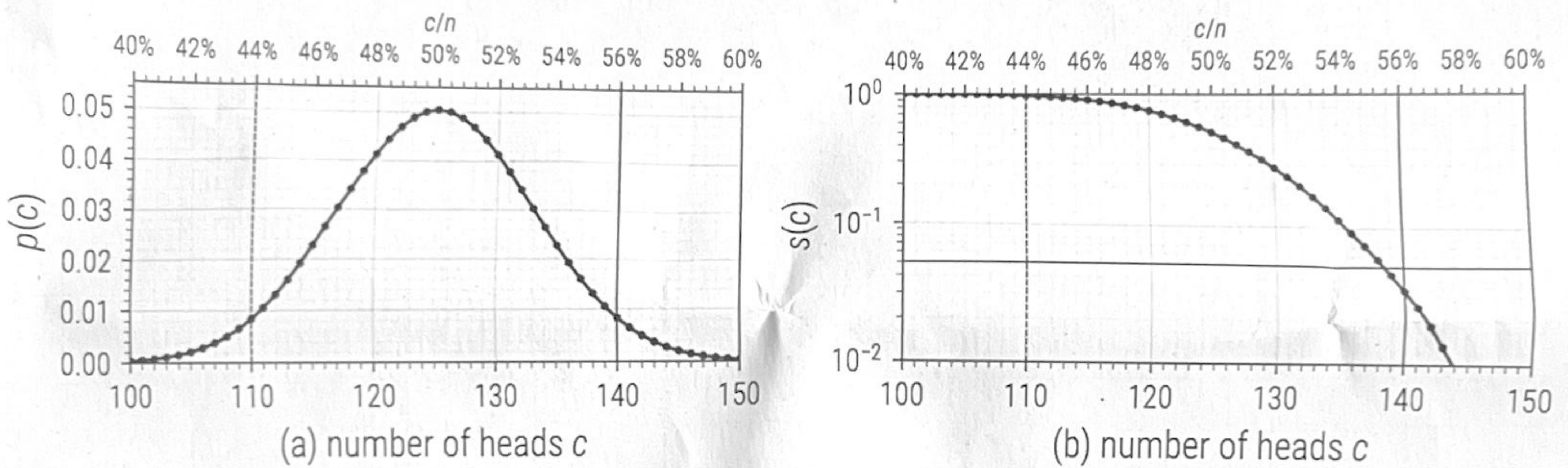
Question 4 of 5: Testing. When Euro coins were first introduced, there was worry about the quality of the new European money. On January 4, 2002, *The Guardian* reported on an experiment by Tomasz Gliszczynski and Wacław Zawadowski, statistics teachers at the Akademia Podlaska in Siedlce (Poland doesn't have the Euro, but they had received Belgian Euro coins from Poles returning from jobs in Belgium):

When spun on edge 250 times, a Belgian one-euro coin came up heads 140 times and tails 110. 'It looks very suspicious to me', said Barry Blight, a statistics lecturer at the London School of Economics. 'If the coin were unbiased the chance of getting a result as extreme as that would be less than 7%'.
New Coin

The *New Scientist* in a similar report, added

But Howard Grubb, an applied statistician at the University of Reading, notes that, "with a sample of only 250, anything between 43.8 per cent and 56.2 per cent on one side or the other cannot be said to be biased".

Plot (a) below shows the binomial probability mass function for the number c of heads in $n = 250$ coin tosses, assuming a fair coin. Plot (b) shows the survival function $s(c) = 1 - \sum_{c'=0}^{c-1} p(c')$ for the same distribution.



Answer the following questions (extra writing space available overleaf):

1. To the nearest percent, what is the probability of getting 140 heads in 250 tosses of a fair coin? 2 points
2. What is the probability of getting ≥ 140 heads in 250 tosses of a fair coin? 2 points
3. What is the probability of getting ≥ 140 heads or tails in 250 tosses of a fair coin? 2 points
4. What kind of hypothesis is being tested here? Simple or composite? 4 points
5. What kind of test is Grubb referring to? One-sided or two-sided? 4 points
6. What number of heads would be needed to reject the null hypothesis of a fair coin at the 5% significance level? 4 points

Question 5 of 5: Multiple Choice QuestionsWhich of the following statements is **true**?

Note: each question can have multiple correct answers. Each incorrect answer (i.e. a true statement not marked as such, or a false statement marked as true) reduces the count by one point; but no question can yield less than zero points. If you should need to correct one of your answers, make it very clear which is your intended answer. Unclearly labeled answers count as incorrect answers.

1. Probability Distributions: Consider the probability distribution characterised by the discrete probability mass function $p(x, y)$ for the random variables $x, y \in D := \{0, 1, 2, 3\}$. 4 points
 - ☐ It holds that $\sum_{x \in D} p(x | y) = 1$
 - ☐ It holds that $\sum_{y \in D} p(x, y) = 1$
 - ☐ It holds that $\sum_{x \in D} p(x) = 1$
 - ☐ It holds that $\sum_{y \in D} p(x | y) = 1$
2. Properties of random variables: Consider the probability density function $p(x)$ of the random variable $x \in \mathbb{R}$. 4 points
 - ☐ The mean of x is given by $\mathbb{E}_p[x] = \int_{-\infty}^{\infty} xp(x) dx$
 - ☐ The standard deviation of x is given by $\int_{-\infty}^{\infty} \sqrt{x - x^2} p(x) dx$
 - ☐ The variance of x is given by $\int_{-\infty}^{\infty} (x - \mathbb{E}_p[x])^2 p(x) dx$
 - ☐ The variance of x is given by $\int_{-\infty}^{\infty} x^2 p(x) dx$
3. Estimators. Consider a dataset $\mathbf{x} = \{x_1, \dots, x_N\}$ assumed to be drawn iid. from the distribution with probability density function $p(x)$, and the function $\hat{m} = \frac{1}{N} \sum_{n=1}^N x_n$. 4 points
 - ☐ The variance of $\hat{m}$ is given by $\int_{-\infty}^{\infty} (x - \mathbb{E}_p[x])^2 p(x) dx$
 - ☐ $\hat{m}$ is an unbiased estimator of the mean of $p(x)$
 - ☐ The standard error of $\hat{m}$ is the standard deviation of $\hat{m}$
 - ☐ The standard error of $\hat{m}$ is the standard deviation of $p(x)$
4. Confidence. Let $\theta \in \mathbb{R}$ be a parameter of the probability distribution $p(x)$ from above, and $C_n = (L_n, U_n)$ a $1 - \alpha$ confidence interval for θ . This means ... 4 points
 - ☐ x lies within C_n with probability $1 - \alpha$
 - ☐ C_n contains at least a fraction $1 - \alpha$ of the points in x
 - ☐ C_n contains the true value of θ with probability $1 - \alpha$
 - ☐ C_n contains the true value of θ with probability α
5. Testing. If we say we can reject the Null hypothesis at the 5% level, this means 4 points
 - ☐ The null hypothesis is true with probability 5%
 - ☐ The null hypothesis is false with probability 95%
 - ☐ If the null hypothesis is true, the probability of observing the data we have observed is 5%
 - ☐ None of the above